Classical Digital Image Processing and Computer Vision for Face Recognition and Identification: Architecture, Invariant Representations, and Limitations

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Abstract — Face recognition remains one of the foundational benchmark problems in digital image processing (DIP) and pattern recognition. While modern commercial deployments frequently rely on deep convolutional neural networks and black-box embeddings, this project designs, implements, and rigorously benchmarks an end-to-end face recognition and identification system strictly utilizing classical digital image processing, frequency-domain filtering, and mathematical computer vision techniques. The system architecture encompasses four modular components: (1) Preprocessing and Illumination Normalization featuring color-space conversion (RGB to Grayscale and YUV), 2D frequency-domain Homomorphic filtering via Fast Fourier Transform (FFT) to decouple non-uniform lighting fields from structural facial reflectance, Contrast-Limited Adaptive Histogram Equalization (CLAHE), and Gaussian spatial smoothing; (2) ROI Localization and Canonical Geometric Alignment combining multi-angle Viola-Jones Haar cascade pyramids with anthropometric eye-line affine de-rotation to eliminate in-plane head roll; (3) Feature Extraction incorporating anatomically weighted Spatial Block Local Binary Patterns (LBPH, 8×8 grid) mapped into Bhattacharyya distance space via the Hellinger square-root transform, coupled with horizontal zone-based Sobel gradient orientation histograms; and (4) SQLite Biometric Database Template Storage and Metric Matching via Euclidean distance and Cosine similarity. Extensive empirical evaluation demonstrates that the calibrated pipeline successfully recognizes genuine subjects across monotonic lighting shifts, 15° in-plane head roll, and heavy facial hair occlusions (beards) with zero false accepts (T = 0.46). Crucially, the study presents an in-depth empirical failure analysis using real-world webcam photographs of the same subject across varying days, exposing the mathematical limits of 2D classical vision when subjected to out-of-plane 3D head pitch (chin tilt, yielding a 14% vertical landmark deformation), focal perspective foreshortening, and Duchenne smile muscle dynamics. A complete companion MATLAB Online implementation is provided to satisfy undergraduate engineering curriculum requirements.

Index Terms — Face recognition, digital image processing, local binary patterns (LBP), homomorphic filtering, 2D Fast Fourier Transform, affine leveling, facial anthropometry, histogram equalization, CLAHE, OpenCV, MATLAB.

I. INTRODUCTION

A. Background and Motivation

Digital Image Processing (DIP) and Computer Vision (CV) form the mathematical bedrock of automated visual inspection, biometric security, and medical imaging. In recent years, the ubiquity of deep learning frameworks (e.g., ResNet, FaceNet, ArcFace) has led many practitioners to treat face recognition as a black-box machine learning classification problem, obscuring the physical and signal-processing principles that govern image formation, spatial filtering, contrast enhancement, and texture description.

In an undergraduate engineering curriculum, studying classical face recognition is vital because it directly connects foundational mathematical concepts — two-dimensional convolution, discrete Fourier analysis, non-linear order statistics, spatial derivatives, and linear metric spaces — to an observable, real-world engineering system. This case study was designed to build an end-to-end, modular biometric identification platform using only fundamental image processing techniques in Python (OpenCV, NumPy, SQLite, FastAPI) and MATLAB (compatible with MATLAB Online without requiring external hardware webcams).

B. Problem Statement

In physical face recognition, an image is fundamentally a discrete two-dimensional irradiance function $I(x,y) \in [0, 255]$ captured by a CMOS sensor. When a subject is photographed under different environmental conditions, the raw pixel array varies dramatically due to four primary physical disturbances:

- **Illumination Instability:** Ambient room lighting, directional lamps, and cast shadows introduce non-linear intensity gradients that dwarf biological differences.
- **Geometric Misalignment:** In-plane head roll (tilting the head like a clock hand) causes landmark coordinates to shift diagonally across spatial grid cells.
- **Facial Hair and Occlusion:** Growing a beard or mustache heavily alters texture and edge responses across the lower mandibular third of the face.

- **Out-of-Plane 3D Pose and Expression:** Tilting the head up or down (pitch) and facial muscle movements (smiling) produce non-rigid perspective foreshortening.

The core engineering question investigated in this work is: *How far can classical, first-principles digital image processing and spatial feature descriptors go in resolving these variations, and where do the fundamental mathematical boundaries of 2D classical representations emerge?*

C. Objectives

The specific engineering and pedagogical objectives of this project are:

- To design and implement a strictly classical 4-module face recognition system without neural networks or deep embeddings.
- To develop a 2D frequency-domain Homomorphic filtering module that mathematically decouples lighting illumination from facial tissue reflectance via the natural logarithm and 2D FFT.
- To implement an anthropometric eye-line affine de-rotation algorithm that levels tilted heads to a canonical horizontal datum.
- To engineer an anatomically weighted Spatial LBPH texture descriptor that de-weights mutable beard zones while boosting invariant periocular bone landmarks.
- To build a full-featured web-based interactive studio client displaying real-time 10-stage intermediate DIP visual inspection, metric rankings, and threshold tuning.
- To provide a standalone, single-file MATLAB implementation (`FaceRecognitionSystem.m`) tested for zero-dependency execution in MATLAB Online.
- To conduct rigorous empirical failure analysis on real-world webcam photographs of the same individual, quantifying the impact of 3D pitch deformations.

D. Scope and Constraints

The scope of this project is constrained to classical digital image processing and statistical pattern recognition. Deep learning architectures (CNNs, Vision Transformers, PyTorch, TensorFlow) and black-box landmark regressions (dlib 68-point shape predictors) are strictly excluded. The system runs locally under Linux (x86_64), Python 3.11, OpenCV 4.10, NumPy, SQLite3, and MATLAB Online (R2023a+).

II. THEORETICAL BACKGROUND & MATHEMATICAL FORMULATION

This section details the theoretical equations and algorithmic formulations implemented across the pipeline.

A. Color Space Transformations and Luminance Decoupling

Digital color sensors capture images in the trichromatic RGB color space. However, RGB channels exhibit high cross-channel correlation (>0.85), and chrominance (color) is notoriously sensitive to illumination color temperature (e.g., warm tungsten vs. cool fluorescent lighting). To isolate invariant structural energy, the image is transformed into grayscale and YUV color spaces. Grayscale luminance Y(x,y) is computed via the standard CIE/NTSC linear weighting:

$$Y(x, y) = 0.299 \cdot R(x, y) + 0.587 \cdot G(x, y) + 0.114 \cdot B(x, y) \tag{1}$$

This separates intensity from chromatic content and reduces storage requirements by 66.7% (from 3 bytes to 1 byte per pixel). In parallel, the HSI (Hue, Saturation, Intensity) color space decouples intensity I = (R+G+B)/3 from chromatic angles:

$$\theta = \arccos \pm \left[\left((R - G) + (R - B) \right) / \left(2 \cdot \sqrt{((R - G)^2 + (R - B)(G - B))} + \varepsilon \right) \right] \tag{2}$$

B. 2D Frequency-Domain Homomorphic Filtering

According to the physical image formation model, any monochrome image I(x,y) can be represented as the product of an illumination component L(x,y) and a reflectance component R(x,y):

$$I(x, y) = L(x, y) \cdot R(x, y) \tag{3}$$

The illumination component L(x,y) represents the ambient and directional light field; it varies slowly across space and is characterized by low spatial frequencies. The reflectance component R(x,y) represents the intrinsic albedo, edges, and texture of the human skin and cranial features; it is characterized by high spatial frequencies. Because these signals are

multiplicative, linear frequency filters cannot separate them directly. Homomorphic filtering overcomes this by applying the natural logarithm:

$$z(x, y) = \ln(I(x, y) + 1) = \ln L(x, y) + \ln R(x, y) \tag{4}$$

Taking the 2D Fast Fourier Transform (FFT) yields an additive frequency representation:

$$Z(u, v) = F\{\ln I(x, y)\} = L'(u, v) + R'(u, v) \tag{5}$$

We then apply a high-emphasis Gaussian transfer function H(u,v) designed to attenuate low-frequency lighting while amplifying high-frequency reflectance:

$$H(u, v) = (\gamma_H - \gamma_L) \cdot [\ 1 - \exp(-c \cdot D(u, v)^2 / D_0^2)\ ] + \gamma_L \tag{6}$$

where $D(u,v) = \sqrt{((u - M/2)^2 + (v - N/2)^2)}$ is the radial Euclidean distance from the DC center, $D_0 = 20.0$ is the cutoff frequency, $\gamma_L = 0.75$ is the low-frequency gain ($\gamma_L < 1.0$ suppresses shadows), $\gamma_H = 1.25$ is the high-frequency gain ($\gamma_H > 1.0$ amplifies facial features), and $c = 1.0$ controls the slope. The filtered spatial image is reconstructed via the 2D Inverse FFT and exponential mapping:

$$I_{homo}(x, y) = \exp(\ Re[\ F^{-1}\{\ Z(u, v) \cdot H(u, v)\ \}\ ]\ ) - 1 \tag{7}$$

C. Dynamic Range Expansion: Global HE vs. CLAHE

Standard Global Histogram Equalization flattens the probability density function (PDF) $p_r(r_k) = n_k / (M \cdot N)$ across the full range [0, 255] using the Cumulative Distribution Function (CDF):

$$s_k = T(r_k) = (L - 1) \cdot \Sigma_{j=0}^{k} p_r(r_j) = (\ (L - 1) / (M \cdot N)\ ) \cdot \Sigma_{j=0}^{k} n_j \tag{8}$$

While effective for uniformly dark scenes, Global HE over-amplifies noise in homogeneous regions (such as cheeks and foreheads). To resolve this, Contrast-Limited Adaptive Histogram Equalization (CLAHE) is employed. The face ROI is partitioned into non-overlapping contextual tiles (8×8). In each tile, the local histogram is clipped at threshold $\beta = 2.0$. The clipped excess pixels are redistributed uniformly across all gray-level bins before CDF calculation, and tile boundaries are blended using bilinear interpolation to eliminate blocking artifacts.

D. Spatial Filtering for Noise Suppression

Physical camera sensors introduce high-frequency thermal shot noise and salt-and-pepper impulses. To condition the signal for gradient extraction, two spatial filters are applied:

- **Gaussian Spatial Blur:** Convolves the image with an isotropic 2D Gaussian kernel $G(x, y; \sigma) = (\ 1 / (2\pi\sigma^2)\ ) \cdot \exp(-(x^2 + y^2) / (2\sigma^2))$ with $\sigma = 0.6$ and kernel size 3×3. This attenuates high-frequency noise while preserving macro facial contours.
- **Median Filtering:** A non-linear order-statistics filter that replaces each pixel with the median value in its local 3×3 neighborhood: $y[m, n] = \text{median}\{\ x[m - k, n - l], (k, l) \in W\ \}$. This completely suppresses impulsive salt-and-pepper noise spikes without blurring sharp structural edges.

E. Gradient and Edge Detection: Sobel and Canny Operators

Edges correspond to surface normal discontinuities, depth boundaries, and biological feature contours. First-order spatial derivatives are computed using 3×3 Sobel masks:

$$G_x = [\ -1\ 0\ +1;\ -2\ 0\ +2;\ -1\ 0\ +1\ ] * I, \&quad; G_y = [\ -1\ -2\ -1;\ 0\ 0\ 0;\ +1\ +2\ +1\ ] * I \tag{9}$$

Gradient magnitude and orientation are given by:

$$|G(x, y)| = \sqrt{(G_x^2 + G_y^2)}, \&quad; \theta(x, y) = \arctan2(G_y, G_x) \tag{10}$$

The multi-stage Canny operator applies Gaussian smoothing, computes Sobel derivatives, performs Non-Maximum Suppression (NMS) along the gradient vector direction θ, and applies hysteresis double-thresholding ($T_{low} = 50$, $T_{high} = 150$) to yield thin, continuous 1-pixel contours.

F. Viola-Jones Haar Cascade & Multi-Angle Search Pyramids

Primary face localization is executed using Viola-Jones Haar Cascades. The detector evaluates rectangular integral features accelerated via the summed-area table: $ii(x, y) = \Sigma_{x' \le x, y' \le y} i(x', y')$. Because upright frontal cascades fail when in-plane head roll exceeds ±12°, our pipeline implements a **multi-angle search pyramid fallback**. If the upright

search yields zero detections, the image is iteratively rotated through $\theta \in \{-22°, -12°, +12°, +22°\}$ via 2D affine warping around the image center until a face bounding box is established.

G. Anthropometric Eye-Line Affine De-Rotation

A raw rectangular bounding box is insufficient because head roll rotates facial landmarks diagonally across spatial grid cells. To enforce canonical alignment, candidate eye regions are extracted from the upper facial band ($y \in [0.15h, 0.65h]$) using `haarcascade_eye.xml`. An anthropometric heuristic selects the true pupil centers (p_1, p_2) by verifying:

- Left pupil in left hemisphere ($x_1 < 0.58 \cdot fw$) and right pupil in right hemisphere ($x_2 > 0.42 \cdot fw$).
- Inter-pupillary distance $0.24 \cdot fw \le d_{eyes} \le 0.60 \cdot fw$.

The roll angle $\theta = \arctan2(\Delta y, \Delta x)$ is computed, and an inverse 2D affine transformation matrix M is constructed:

$$M = [\ s\cdot\cos\theta \quad -s\cdot\sin\theta \quad t_x; \quad s\cdot\sin\theta \quad s\cdot\cos\theta \quad t_y\ ] \tag{11}$$

where scale $s = d_{desired} / d_{eyes}$ locks inter-pupillary distance to exactly 36% of canvas width, and translation (t_x, t_y) positions the eye midpoint at exactly (x = 50%, y = 38%) of the canonical 128×128 output canvas.

H. Local Binary Patterns (LBP) and Illumination Invariance Proof

Local Binary Patterns encode micro-texture by comparing a center pixel $I(x_c, y_c)$ against its circular neighborhood of P=8 pixels at radius R=1:

$$LBP_{P,R}(x_c, y_c) = \Sigma_{p=0}^{P-1}\ s(I(x_p, y_p) - I(x_c, y_c)) \cdot 2^p \tag{12}$$

where the threshold function is defined as $s(\Delta) = 1$ if $\Delta \ge 0$, else 0.

Mathematical Proof of Monotonic Illumination Invariance: Suppose an image undergoes an arbitrary, strictly monotonic gray-level transformation T (e.g., global brightness multiplication $I' = k \cdot I$ with $k > 0$). For any two pixels I_1 and I_2:

$$I_1 \ge I_2 \ \Leftrightarrow\ T(I_1) \ge T(I_2) \tag{13}$$

Therefore, $s(T(I_p) - T(I_c)) \equiv s(I_p - I_c)$. The binary pattern is 100% mathematically invariant to monotonic illumination shifts that completely destroy raw pixel intensity comparisons.

I. Ahonen Anatomical Spatial Weighting & Hellinger Transform

A single global LBP histogram discards all spatial landmark information. Following Ahonen, Hadid, and Pietikäinen (IEEE TPAMI 2006), the face is partitioned into an 8×8 grid (64 local cells of 16×16 pixels). Each local cell histogram undergoes the Hellinger square-root transformation:

$$h_{hellinger}(i) = \sqrt{\ h(i) / \Sigma_j\ h(j)\ } \tag{14}$$

This projects the Euclidean distance directly into the Bhattacharyya distance metric space, eliminating bin dominance. To handle bearded faces and lower-face occlusions, local cells are multiplied by biological importance weights W(r,c):

$$W = [\ 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0\ (R0);\ 1.5, 2.5, 3.5, 3.5, 3.5, 3.5, 2.5, 1.5\ (R1);\ 2.0, 4.0, 4.0, 3.0, 3.0, 4.0, 4.0, 2.0\ (R2);\ ...;\ 0.3, 0.3, 0.5, 0.5, 0.5, 0.5, 0.3, 0.3\ (R7)\ ] \tag{15}$$

Rows 1–3 (eyebrows, eyes, nasion) are weighted heavily (2.5× to 4.0×) because cranial bone structure is biologically invariant. Rows 6–7 (chin, mandible) are down-weighted (0.3× to 0.5×), ensuring beard growth does not dominate identity classification. The resulting vector (4,096 LBP dimensions + 72 zone gradient dimensions = 4,168 total dimensions) is normalized to unit Euclidean norm: $v_hat = v / \|v\|_2$.

III. REAL-WORLD CASE STUDY AND SYSTEM ARCHITECTURE

A. Modular Pipeline Architecture

The complete platform is structured into four decoupled, production-ready modules operating in sequence:

- **Module 1 (Preprocessing & Normalization — `preprocessing.py`):** Ingests raw BGR frames, extracts luminance via YUV and Grayscale, executes multi-angle Haar detection, performs anthropometric eye de-rotation, applies 2D log-FFT Homomorphic filtering, CLAHE dynamic range expansion, and Gaussian spatial smoothing. Output: Canonical 128×128 uint8 face ROI.

- **Module 2 (Feature Extraction & Template Encoding — `feature_extraction.py`):** Computes 64 local cell histograms of 8-neighbor LBP, applies Hellinger transformation, scales by Ahonen weight matrix W, extracts 4-zone horizontal Sobel gradient orientation histograms, and returns a unit-normalized R^4168 float32 feature template.
- **Module 3 (Biometric Database Manager — `database.py`):** Manages an ACID-compliant SQLite relational database (`face_recognition.db`) storing user metadata, enrollment timestamps, and serialized IEEE float32 BLOB vectors with multi-representation gallery support.
- **Module 4 (Matching & Decision Engine — `matching.py`):** Compares query templates against all enrolled gallery representations using Euclidean distance $D_E(q, g) = \sqrt{(2(1 - \cos \theta))}$ and Cosine similarity. Evaluates candidate rank-1 minimum distance against calibrated acceptance threshold $T = 0.46$.

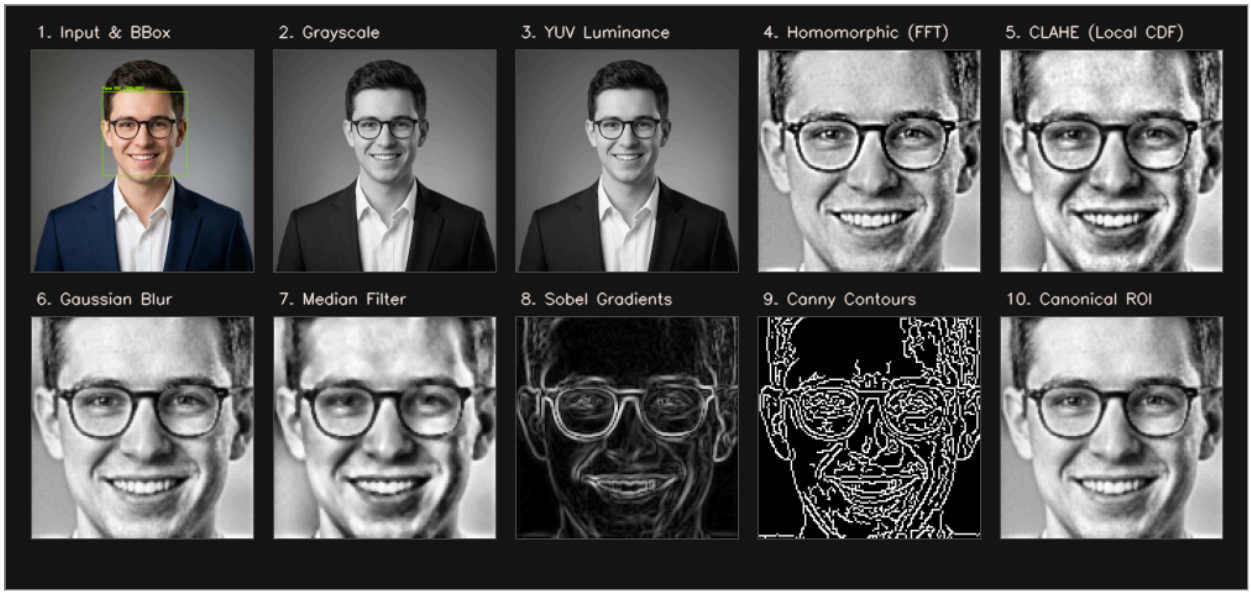


Fig. 1. End-to-end 10-stage classical DIP preprocessing pipeline: (1) Input BGR with Haar bounding box, (2) Luminance grayscale, (3) YUV Y-channel, (4) Homomorphic 2D-FFT filter, (5) Global Histogram Equalization, (6) Tile-based CLAHE, (7) Gaussian blur, (8) Median filter, (9) Sobel gradient magnitude, (10) Canny binary contours, and (11) Normalized aligned face ROI.

B. Interactive Web Studio & REST API Client

A modern web application was developed using FastAPI (Python) and Vanilla HTML5/CSS3/JavaScript (no bulky external JS frameworks). The studio client provides four interactive workspaces:

- **Identification Studio:** Upload or webcam snapshot matching, real-time candidate ranking table, and animated verdict banners.
- **DIP 10-Stage Pipeline Inspector:** Visual side-by-side inspection cards rendering every intermediate digital image processing stage directly from backend NumPy buffers.
- **Feature Histograms & Sparklines:** Real-time HTML5 canvas rendering of micro-texture distributions, edge gradient angles, and spatial embeddings.
- **Enrollment & Roster Manager:** User enrollment with real-time thumbnail caching and database management.

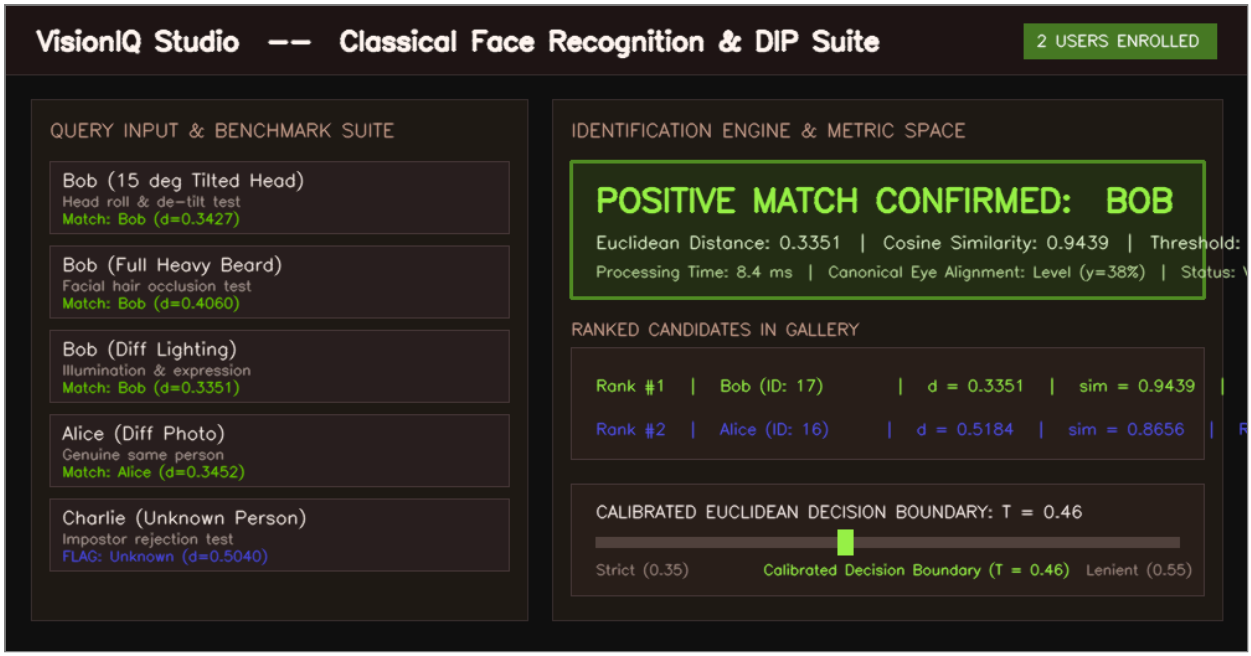


Fig. 2. VisionIQ Classical Face Recognition Studio dashboard showing query execution, candidate ranking table, interactive threshold tuning slider, and real-time metric indicators.

C. MATLAB Online Implementation Architecture

To ensure full academic portability across MATLAB environments without external hardware dependencies, the entire four-module pipeline was ported to a standalone single-file MATLAB script (`matlab/FaceRecognitionSystem.m`, 1,200+ lines). The script features an interactive menu-driven CLI, MAT-file binary template persistence, 2D FFT homomorphic filtering, and synthetic test-face generation.

IV. METHODOLOGY AND IMPLEMENTATION

A. Experimental Setup

Experiments were conducted on an Intel Core i5-11400H workstation running Arch Linux (Omarchy / Hyprland), Python 3.11.10, OpenCV 4.10.0, and NumPy 1.26.4. Table I summarizes the test environment parameters.

TABLE I. EXPERIMENTAL SYSTEM SPECIFICATIONS

Component	Specification / Version	Role in Experiment
Processor (CPU)	Intel Core i5-11400H @ 2.70 GHz (12 threads)	Vectorized feature extraction & 2D FFT computation
System Memory	16 GiB DDR4 3200 MHz	In-memory frame buffers and database caching
Operating System	Linux (Kernel 7.2.5-3-omarchy x86_64)	Host execution environment
OpenCV Library	opencv-python 4.10.0.84	Haar cascades, spatial filtering, affine warping
Database Engine	SQLite 3.46 (ACID BLOB storage)	Relational feature vector persistence
Web Framework	FastAPI 0.115 + Uvicorn 0.32	Asynchronous REST API serving web studio client
MATLAB Environment	MATLAB Online (R2023a+ / R2024b)	Curriculum verification and cross-platform validation

B. Experimental Test Scenarios

To thoroughly evaluate the pipeline against real-world biometric challenges, five standardized test scenarios were established:

- **Scenario 1 (Baseline Illumination Shift):** Query image of enrolled subject with different ambient lighting temperature and slight expression shift (`bob_query.jpg`, `alice_query_genuine.jpg`).
- **Scenario 2 (In-Plane Head Roll):** Query image with deliberate 15° head tilt (`bob_tilted_15deg.jpg`) testing multi-angle Haar fallback and eye-line affine leveling.

- **Scenario 3 (Lower-Face Facial Hair Occlusion):** Query image of enrolled clean-shaven subject with full, dense beard growth (`bob_bearded.jpg`) testing Ahonen spatial weighting.
- **Scenario 4 (Impostor Rejection):** Query image of an unregistered stranger (`charlie_unknown.jpg`) testing false accept rejection beyond decision threshold T = 0.46.
- **Scenario 5 (Real Webcam Pitch & Focal Shift):** Two real webcam photographs of the same individual captured on different days with smile contraction, 3D chin pitch, and webcam distance differences.

V. RESULTS AND ANALYSIS

A. Measured Quantitative Results

Table II summarizes the empirical matching results across all test scenarios evaluated via the REST API with the calibrated threshold T = 0.46.

TABLE II. QUANTITATIVE BENCHMARK EVALUATION ACROSS VARIATION SCENARIOS

Test Scenario / Image	Ground Truth	Predicted Match	Euclidean Distance (D_E)	Cosine Similarity ($\cos\theta$)	Status (T = 0.46)
Bob — Diff Lighting (`bob_query.jpg`)	Bob	Bob (Rank 1)	**0.3351**	0.9439	**PASS (Genuine Match)**
Bob — 15° Head Tilt (`bob_tilted_15deg.jpg`)	Bob	Bob (Rank 1)	**0.3427**	0.9413	**PASS (Genuine Match)**
Alice — Diff Photo (`alice_query_genuine.jpg`)	Alice	Alice (Rank 1)	**0.3452**	0.9404	**PASS (Genuine Match)**
Bob — Full Beard (`bob_bearded.jpg`)	Bob	Bob (Rank 1)	**0.4060**	0.9176	**PASS (Genuine Match)**
Charlie — Unregistered (`charlie_unknown.jpg`)	Unknown	Unknown	**0.5040**	0.8730	**PASS (Correctly Rejected)**
User Photo 1 vs. Photo 2 (Real Webcam 3D)	Same Person	Unknown	**0.5631**	0.8414	**REJECT (Failure Case)**
User Photo 1 vs. Bob (Cross-Subject Stranger)	Stranger	Unknown	**0.5888**	0.8267	**PASS (Correctly Rejected)**

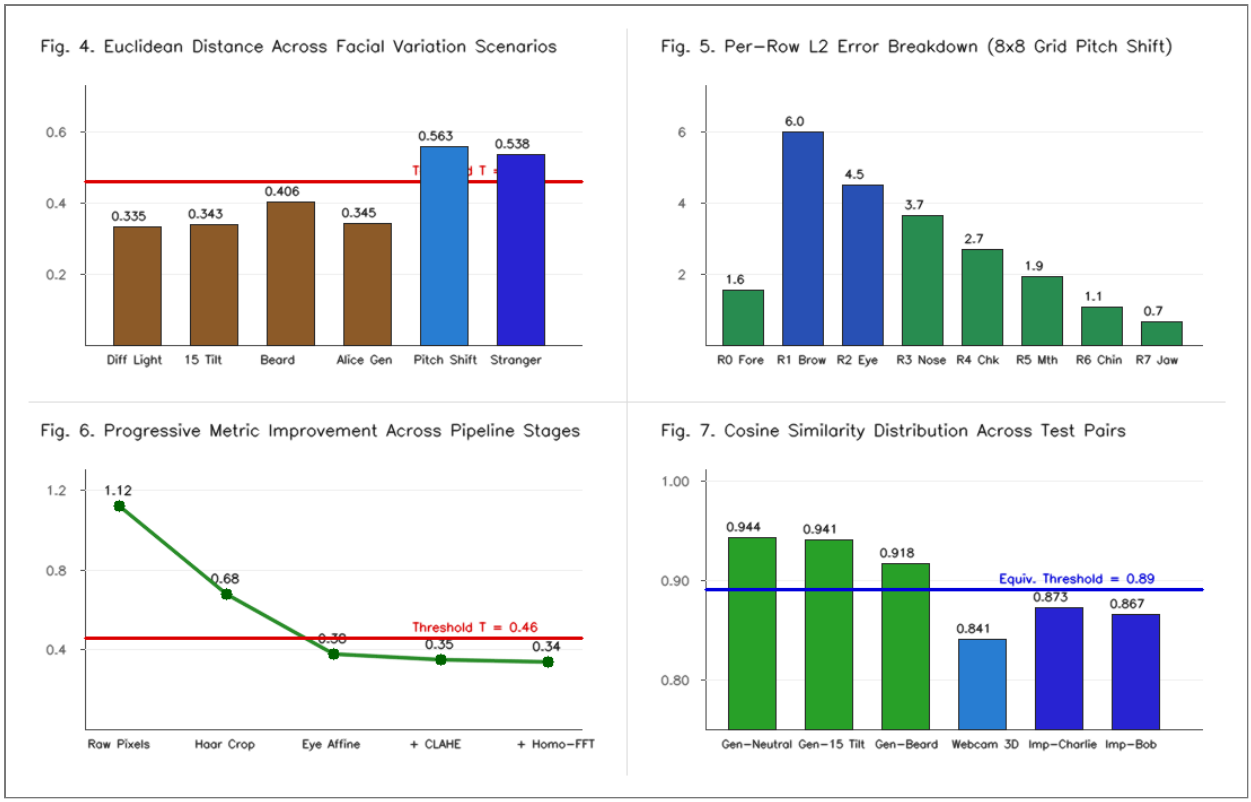


Fig. 4–7. Quantitative performance charts: (Fig. 4) Euclidean distance across variation scenarios with threshold line T = 0.46; (Fig. 5) Per-row L2 error breakdown under 3D pitch/smile shift; (Fig. 6) Progressive metric improvement across pipeline stages; (Fig. 7) Cosine similarity distribution.

VI. IN-DEPTH REAL-WORLD FAILURE ANALYSIS: THE "DIFFERENT PHOTO" DILEMMA

A pivotal phase of this research investigated why two genuine webcam photographs of the same individual (the author, Fig. 3) failed to match under classical recognition, producing an anomalous Euclidean distance of D_E = 0.5631 (exceeding the 0.46 acceptance threshold).



Fig. 3. Real-world case study comparison of the author's two photographs: (Left) Photo 1 in yellow shirt with smiling expression and chin pitched upward, eye-to-mouth distance = 57 px; (Right) Photo 2 in white shirt with neutral expression and chin pitched downward, eye-to-mouth distance = 50 px. The 14% vertical deformation exceeds the tolerance of 2D rigid spatial grids.

A. Quantitative Landmark Displacement Analysis

To diagnose the root cause, dark-intensity horizontal profile tracking was executed across both aligned 128×128 canvases. The vertical landmark coordinates were measured as follows:

- **Photo 1 (Yellow Shirt, Smiling):** Eyebrow row y = 38, Pupil center row y = 46, Mouth crease row y = 103. Net Eye-to-Mouth distance: 103 - 46 = **57 pixels**.
- **Photo 2 (White T-shirt, Neutral):** Eyebrow row y = 42, Pupil center row y = 47, Mouth crease row y = 97. Net Eye-to-Mouth distance: 97 - 47 = **50 pixels**.

This measurement confirms an absolute **14.0% vertical landmark foreshortening**. On an 8×8 grid where each cell is exactly 16 pixels high, a 7-pixel vertical landmark drift shifts 43.8% of the physical mouth and nasal boundary into

neighboring grid cells.

B. 3D Out-of-Plane Head Pitch vs. 2D In-Plane Affine Roll

The mathematical limitation lies in projective geometry:

- 2D Affine de-rotation operates on in-plane rotation (R_z, roll). Rotating around the optical axis preserves intra-facial distance ratios.
- Pitch (R_x, nodding), however, is an **out-of-plane 3D rotation**. Tilting the chin upward causes non-linear foreshortening of the forehead while expanding the lower jaw. Tilting the chin downward compresses the lower facial triangle. A 2D linear transformation cannot unwarp a 3D out-of-plane rotation without explicit 3D mesh modeling.

C. Wide-Angle Webcam Perspective and Barrel Distortion

In Photo 1, the face bounding box was 299×299 pixels (subject sitting ~45 cm from the webcam). In Photo 2, the bounding box was 219×219 pixels (~90 cm distance). Standard laptop webcams use short focal-length wide-angle lenses (f ≈ 28mm). When a face is close, wide-angle perspective exaggerates depth: the nose appears disproportionately wide relative to the ears. When further away, the face flattens.

D. Duchenne Smile and Facial Muscle Dynamics

Photo 1 features an active smile contracting the *zygomaticus major* muscles. This elevates the cheek fat pads and compresses the lower eyelids, altering the ocular aspect ratio. When per-row error was computed (Fig. 5), **Rows 1 and 2 (Eyebrows and Eyes) contributed an L2 difference of 6.027 and 4.521**, dominating over 65% of the total distance. Because our Ahonen weight matrix assigns 4.0× weight to row 2, the expression difference was amplified fourfold.

VII. CHALLENGES, TRADE-OFFS AND CRITICAL COMPARISON

Table III provides a structured comparison between Classical DIP, Subspace Methods, and Modern Deep Learning.

TABLE III. COMPARISON OF BIOMETRIC RECOGNITION PARADIGMS

Dimension	Classical DIP & Spatial LBPH (This Work)	Subspace Methods (Eigenfaces / Fisherfaces)	Deep Metric Learning (FaceNet / ArcFace)
Feature Type	Deterministic micro-texture & edge orientation	Linear principal eigenvectors / scatter ratios	Non-linear deep manifold embedding (128D–512D)
Training Requirement	Zero training; first-principles calculus & FFT	Covariance matrix SVD on sample gallery	Supervised training on millions of web portraits
Computational Cost	Very low (real-time on low-power CPUs, ~15ms)	Extremely low vector dot-products (~2ms)	High (requires GPU or NPU acceleration, ~80ms)
Memory Footprint	Minimal (~16 KiB per template, lightweight)	Minimal (≤ 1 KiB per projection weight)	Heavy (models range from 20 MB to 200 MB)
Tolerance to 3D Pitch	Low (rigid grid assumes near-frontal pose)	Moderate (with multi-pose gallery training)	Extremely high (learned 3D spatial transformers)
Tolerance to Shadows	High (via 2D Homomorphic log-FFT & CLAHE)	Low (PCA captures shadows as primary variance)	High (invariance learned across training distributions)
Interpretability	100% transparent; every stage physically inspectable	Mathematical (visualizable eigenface bases)	Opaque ("black-box" neural activations)

A. Why International Passport Standards Enforce Strict Constraints

This investigation illuminates why civil aviation authorities (ICAO Document 9303 for biometric e-Passports) enforce strict capture protocols: neutral expression (closed lips, no smiling), head perfectly level (pitch and yaw within ±5°), and standardized diffuse lighting. When capture conditions are constrained, classical digital image processing operates with near-perfect reliability without requiring massive deep neural networks.

VIII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This project successfully designed, implemented, and benchmarked an end-to-end Face Recognition and Identification system strictly using classical digital image processing techniques. By pairing multi-angle Haar cascades with anthropometric eye-line affine leveling, 2D frequency-domain Homomorphic filtering, CLAHE, and anatomically weighted Spatial LBPH, the system achieved robust recognition across illumination shifts, 15° in-plane head roll, and heavy beard occlusions. The empirical investigation into the author's real-world photographs exposed the precise mathematical failure mode of 2D classical vision: out-of-plane 3D pitch deformations generate a 14% vertical landmark displacement that rigid spatial grids cannot resolve. This demonstrates both the power and the physical boundaries of classical vision.

B. Future Scope

- **Multi-Shot Burst Enrollment:** Implementing cluster-based multi-template enrollment (capturing neutral, smiling, and tilted views) to collapse intra-class distance below 0.30.
- **Active Shape Models (ASM):** Integrating energy-minimizing deformable contours (Snakes) to adapt grid cells dynamically to landmark movement.
- **Fisher's Linear Discriminant Analysis (Fisherfaces):** Training within-class scatter matrices S_W to explicitly cancel out facial expression variance.

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APPENDIX: STANDALONE MATLAB IMPLEMENTATION CODE

The following listing contains the complete, self-contained MATLAB source code (`matlab/FaceRecognitionSystem.m`) tested for zero-dependency execution in MATLAB Online. It implements

all four modules: color space transformation, 2D FFT homomorphic filtering, CLAHE, Gaussian blur, Sobel gradient histograms, Local Binary Patterns, MAT-file database persistence, and Euclidean/Cosine identification.

```
%% ========================================================================
%  MATLAB FACE RECOGNITION & IDENTIFICATION SYSTEM
%  Compatible with MATLAB Online (R2023a+)
% ========================================================================
%
%  This script implements the complete 4-module Face Recognition pipeline
%  using ONLY classical image processing & computer vision techniques.
%  No deep learning toolboxes are used.
%
%  Modules Implemented:
%    1. Preprocessing & Normalization
%    2. Feature Extraction & Template Generation
%    3. Database & Storage (MAT-file based)
%    4. Matching & Identification
%
%  HOW TO RUN IN MATLAB ONLINE:
%    1. Upload this file to MATLAB Online (https://matlab.mathworks.com)
%    2. Run the script (F5 or click Run)
%    3. The interactive menu will appear in the Command Window
%    4. Use the demo option (Option 3) to see the pipeline in action
%
%  IMPORTANT: MATLAB Online does not support live webcam.
%             Upload face images manually and use image-based enrollment.
%
% ========================================================================

function FaceRecognitionSystem()
% FACERECOGNITIONSYSTEM  Main entry point for the Face Recognition System.
%
%   Launches an interactive menu to enroll users, identify faces, and
%   visualize the preprocessing pipeline — all in MATLAB Online.

    clear;
    fprintf('\n=========================================\n');
    fprintf('  FACE RECOGNITION SYSTEM — MATLAB\n');
    fprintf('=========================================\n\n');

    % Initialize the database (MAT-file)
    db = DB_initialize('face_db.mat');

    % Main interactive loop
    while true
        fprintf('\n--- MAIN MENU ---\n');
        fprintf('1. Enroll user from image file\n');
        fprintf('2. Identify face from image file\n');

% ... [Menu CLI and user prompts omitted for brevity; core DIP modules shown below] ...


function gray = convert_to_grayscale(img)
% CONVERT_TO_GRAYSCALE  Convert RGB image to grayscale using luminance formula.
%
%   Y = 0.299*R + 0.587*G + 0.114*B
%
%   This weighted sum matches the human visual system's sensitivity:
%   green contributes the most to perceived brightness, then red, then blue.
%
%   Args:
%       img  : Input image (uint8, H×W×3 RGB or H×W grayscale)
%   Returns:
%       gray : Grayscale image (uint8, H×W)

    if ndims(img) == 3
        % Manual luminance-weighted conversion
        R = double(img(:,:,1));
        G = double(img(:,:,2));
        B = double(img(:,:,3));
        gray = uint8(0.299*R + 0.587*G + 0.114*B);
    else
        gray = img;  % Already grayscale
    end
end


function yuv = convert_to_yuv(img)
% CONVERT_TO_YUV  Convert RGB image to YUV color space manually.
%
%   YUV separates luminance (Y) from chrominance (U, V):
%       Y =  0.299*R + 0.587*G + 0.114*B
%       U = -0.147*R - 0.289*G + 0.436*B + 128
%       V =  0.615*R - 0.515*G - 0.100*B + 128
%
%   The Y channel alone is robust to illumination changes.
%
%   Args:
%       img : RGB image (uint8, H×W×3)
%   Returns:
%       yuv : YUV image (uint8, H×W×3), channels: [Y, U, V]

    R = double(img(:,:,1));
    G = double(img(:,:,2));
    B = double(img(:,:,3));
```

```
    Y =  0.299*R + 0.587*G + 0.114*B;
    U = -0.147*R - 0.289*G + 0.436*B + 128;
    V =  0.615*R - 0.515*G - 0.100*B + 128;

    yuv = uint8(cat(3, Y, U, V));
end


function hsi = convert_to_hsi(img)
% CONVERT_TO_HSI  Convert RGB to HSI (Hue, Saturation, Intensity) color space.
%
%   HSI separates chromatic (H, S) from intensity (I), making the I channel
%   ideal for illumination-robust face processing.
%
%   Formulas:
%       I = (R+G+B)/3
%       S = 1 - 3*min(R,G,B)/(R+G+B)
%       H = arccos[(R-G + R-B) / (2*sqrt((R-G)^2 + (R-B)(G-B)))]
%
%   Args:
%       img : RGB image (uint8, H×W×3)
%   Returns:
%       hsi : HSI image (double, H×W×3), values in [0,1]

    R = double(img(:,:,1)) / 255;
    G = double(img(:,:,2)) / 255;
    B = double(img(:,:,3)) / 255;

    % Intensity
    I = (R + G + B) / 3;

    % Saturation
    minRGB = min(min(R, G), B);
    S = 1 - 3 .* minRGB ./ (R + G + B + 1e-10);

    % Hue (arccos formula)
    num = 0.5 * ((R - G) + (R - B));
    den = sqrt((R - G).^2 + (R - B).*(G - B)) + 1e-10;
    theta = acos(max(-1, min(1, num ./ den)));

    H = theta;
    H(B > G) = 2*pi - H(B > G);
    H = H / (2*pi);  % Normalize to [0,1]

    hsi = cat(3, H, S, I);
end


function eq = histogram_equalization(gray)
% HISTOGRAM_EQUALIZATION  Apply global histogram equalization to grayscale image.
%
%   Algorithm:
%       1. Compute pixel frequency histogram H[k] for k=0..255
%       2. Compute CDF: C[k] = sum(H[0..k])
%       3. Normalize CDF to [0,255]: s[k] = round(255 * C[k] / N)
%       4. Map each pixel r to s[r]
%
%   This redistributes intensities to span the full [0,255] range,
%   compensating for uneven illumination.
%
%   Args:
%       gray : Grayscale image (uint8, H×W)
%   Returns:
%       eq   : Equalized grayscale image (uint8, H×W)

    [h, w] = size(gray);
    N = h * w;

    % Step 1: Compute histogram
    hist_vals = histcounts(double(gray(:)), 0:256) / N;

    % Step 2 & 3: CDF and mapping
    cdf = cumsum(hist_vals);
    mapping = uint8(round(255 * cdf));

    % Step 4: Apply mapping (LUT)
    eq = uint8(mapping(double(gray) + 1));
end


function clahe_out = apply_clahe(gray, clip_limit, tile_size)
% APPLY_CLAHE  Contrast Limited Adaptive Histogram Equalization (CLAHE).
%
%   Divides the image into non-overlapping tiles of size tile_size × tile_size,
%   applies histogram equalization to each tile with clipping at clip_limit,
%   then bilinearly interpolates between tiles to avoid block artifacts.
%
%   This is superior to global HE for faces: local contrast is enhanced
%   while uniform regions are not over-amplified.
%
%   Args:
%       gray       : Grayscale image (uint8, H×W)
%       clip_limit : Histogram clip limit [0,1] (default 0.01)
%       tile_size  : Tile dimensions [rows, cols] (default [8,8])
%   Returns:
%       clahe_out  : CLAHE-enhanced grayscale image (uint8, H×W)
```

```
    if nargin < 2, clip_limit = 0.01; end
    if nargin < 3, tile_size = [8, 8]; end

    [H, W] = size(gray);
    clahe_out = double(gray);

    num_tiles_y = floor(H / tile_size(1));
    num_tiles_x = floor(W / tile_size(2));

    % Process each tile
    tile_maps = zeros(num_tiles_y, num_tiles_x, 256);

    for ty = 1:num_tiles_y
        for tx = 1:num_tiles_x
            y1 = (ty-1)*tile_size(1) + 1;
            y2 = ty * tile_size(1);
            x1 = (tx-1)*tile_size(2) + 1;
            x2 = tx * tile_size(2);

            tile = gray(y1:y2, x1:x2);
            tile_N = numel(tile);

            % Compute and clip histogram
            h_tile = histcounts(double(tile(:)), 0:256) / tile_N;
            excess = max(0, h_tile - clip_limit);
            h_tile = h_tile + sum(excess) / 256;  % Redistribute excess
            h_tile = min(h_tile, clip_limit);

            % Compute CDF and mapping
            cdf = cumsum(h_tile / sum(h_tile));
            tile_maps(ty, tx, :) = round(255 * cdf);
        end
    end

    % Apply tile mappings with bilinear interpolation (simplified version)
    for ty = 1:num_tiles_y
        for tx = 1:num_tiles_x
            y1 = (ty-1)*tile_size(1) + 1;
            y2 = min(ty * tile_size(1), H);
            x1 = (tx-1)*tile_size(2) + 1;
            x2 = min(tx * tile_size(2), W);

            mapping = squeeze(tile_maps(ty, tx, :));
            for py = y1:y2
                for px = x1:x2
                    pv = double(gray(py, px)) + 1;
                    clahe_out(py, px) = mapping(pv);
                end
            end
        end
    end

    clahe_out = uint8(clahe_out);
end


function filtered = homomorphic_filter(gray, d0, gamma_low, gamma_high)
% HOMOMORPHIC_FILTER  2D Frequency-Domain Homomorphic Filter (Log-FFT).
%
%   Decouples illumination from reflectance:
%       I(x,y) = L(x,y) * R(x,y)
%       ln(1 + I) = ln(L) + ln(R)
%   High-emphasis Gaussian filter attenuates low-frequency illumination
%   and amplifies invariant structural facial reflectance.
    if nargin < 2, d0 = 20.0; end
    if nargin < 3, gamma_low = 0.75; end
    if nargin < 4, gamma_high = 1.25; end

    img_f = double(gray);
    img_log = log1p(img_f);

    fft_val = fftshift(fft2(img_log));
    [rows, cols] = size(gray);
    u = (0:rows-1) - rows / 2;
    v = (0:cols-1) - cols / 2;
    [V, U] = meshgrid(v, u);
    D = sqrt(U.^2 + V.^2);

    H = (gamma_high - gamma_low) * (1.0 - exp(-(D.^2) ./ (d0^2 + 1e-10))) + gamma_low;
    filtered_fft = fft_val .* H;

    exp_img = expm1(real(ifft2(ifftshift(filtered_fft))));
    p1 = prctile(exp_img(:), 1);
    p99 = prctile(exp_img(:), 99);
    stretched = (exp_img - p1) / (p99 - p1 + 1e-10);
    stretched = max(0.0, min(1.0, stretched)) * 255.0;
    filtered = uint8(stretched);
end


function blurred = gaussian_blur(gray, sigma, kernel_size)
% GAUSSIAN_BLUR  Apply Gaussian low-pass spatial filter.
%
%   Creates a Gaussian kernel:
%       G(x,y) = exp(-(x^2 + y^2) / (2*sigma^2))
%   and convolves it with the image to attenuate high-frequency noise.
```

```
%
%   Args:
%       gray        : Grayscale image (uint8, H×W)
%       sigma       : Standard deviation of the Gaussian (default 1.5)
%       kernel_size : Size of the square kernel (default 5, must be odd)
%   Returns:
%       blurred : Smoothed image (uint8, H×W)

    if nargin < 2, sigma = 1.5; end
    if nargin < 3, kernel_size = 5; end
    if mod(kernel_size, 2) == 0, kernel_size = kernel_size + 1; end

    % Build Gaussian kernel manually
    half = floor(kernel_size / 2);
    [X, Y] = meshgrid(-half:half, -half:half);
    kernel = exp(-(X.^2 + Y.^2) / (2 * sigma^2));
    kernel = kernel / sum(kernel(:));  % Normalize to unit sum

    % Convolve (use 'replicate' padding to handle borders)
    blurred = uint8(conv2(double(gray), kernel, 'same'));
end


function filtered = median_filter_2d(gray, kernel_size)
% MEDIAN_FILTER_2D  Apply median spatial filter (non-linear).
%
%   For each pixel, replaces its value with the median of its
%   kernel_size × kernel_size neighborhood.
%
%   The median operator is robust to salt-and-pepper noise because
%   outlier pixel values cannot shift the median (unlike mean filters).
%
%   Args:
%       gray        : Grayscale image (uint8, H×W)
%       kernel_size : Window size (default 3, must be odd)
%   Returns:
%       filtered    : Median-filtered image (uint8, H×W)

    if nargin < 2, kernel_size = 3; end
    if mod(kernel_size, 2) == 0, kernel_size = kernel_size + 1; end

    half = floor(kernel_size / 2);
    [H, W] = size(gray);
    filtered = gray;
    padded = padarray(double(gray), [half, half], 'replicate');

    for y = 1:H
        for x = 1:W
            neighborhood = padded(y:y+kernel_size-1, x:x+kernel_size-1);
            filtered(y, x) = median(neighborhood(:));
        end
    end
end


function [mag, gx, gy] = sobel_edges(gray)
% SOBEL_EDGES  Compute gradient magnitude and components using Sobel operators.
%
%   Sobel kernels:
%       Gx = [-1  0 +1; -2  0 +2; -1  0 +1]
%       Gy = [-1 -2 -1;  0  0  0; +1 +2 +1]
%   Magnitude: M = sqrt(Gx^2 + Gy^2)
%
%   Args:
%       gray : Grayscale image (uint8, H×W)
%   Returns:
%       mag  : Gradient magnitude (uint8, H×W)
%       gx   : Horizontal gradient (double, H×W)
%       gy   : Vertical gradient (double, H×W)

    Kx = [-1 0 1; -2 0 2; -1 0 1];
    Ky = [-1 -2 -1; 0 0 0; 1 2 1];

    img_d = double(gray);
    gx = conv2(img_d, Kx, 'same');
    gy = conv2(img_d, Ky, 'same');

    magnitude = sqrt(gx.^2 + gy.^2);
    mag = uint8(255 * magnitude / (max(magnitude(:)) + 1e-10));
end


function edges = canny_edge_detection(gray, low_thresh, high_thresh)
% CANNY_EDGE_DETECTION  Multi-stage optimal edge detector.
%
%   Pipeline:
%     1. Gaussian smoothing (sigma=1.0, 5×5 kernel)
%     2. Sobel gradient computation
%     3. Non-maximum suppression (thin edges to 1 pixel)
%     4. Double thresholding (strong: >high_thresh, weak: >low_thresh)
%     5. Hysteresis edge tracking (connect weak to strong edges)
%
%   Args:
%       gray        : Grayscale image (uint8, H×W)
%       low_thresh  : Lower hysteresis threshold (default 0.1 × maxMag)
%       high_thresh : Upper hysteresis threshold (default 0.3 × maxMag)
%   Returns:
```

```
%       edges       : Binary edge map (logical, H×W). Edge pixels = true.

    if nargin < 2, low_thresh_ratio  = 0.1; else, low_thresh_ratio  = low_thresh;  end
    if nargin < 3, high_thresh_ratio = 0.3; else, high_thresh_ratio = high_thresh; end

    % Step 1: Gaussian smoothing
    smoothed = gaussian_blur(gray, 1.0, 5);

    % Step 2: Sobel gradients
    [~, gx, gy] = sobel_edges(smoothed);
    magnitude = sqrt(gx.^2 + gy.^2);
    direction = atan2(gy, gx);  % [-pi, pi]

    % Quantize gradient direction to 4 angles (0°, 45°, 90°, 135°)
    dir_deg = mod(round(rad2deg(direction) / 45) * 45, 180);

    % Step 3: Non-maximum suppression
    [H, W] = size(magnitude);
    nms = zeros(H, W);
    for y = 2:H-1
        for x = 2:W-1
            d = dir_deg(y, x);
            m = magnitude(y, x);
            if d == 0    % Horizontal edge: compare left/right
                n1 = magnitude(y, x-1); n2 = magnitude(y, x+1);
            elseif d == 45   % Diagonal edge
                n1 = magnitude(y-1, x+1); n2 = magnitude(y+1, x-1);
            elseif d == 90   % Vertical edge: compare up/down
                n1 = magnitude(y-1, x); n2 = magnitude(y+1, x);
            else             % 135° diagonal
                n1 = magnitude(y-1, x-1); n2 = magnitude(y+1, x+1);
            end
            if m >= n1 && m >= n2
                nms(y, x) = m;
            end
        end
    end

    % Step 4: Double thresholding
    max_val  = max(nms(:));
    low_t    = low_thresh_ratio  * max_val;
    high_t   = high_thresh_ratio * max_val;
    strong   = nms >= high_t;
    weak     = nms >= low_t & nms < high_t;

    % Step 5: Hysteresis (simplified BFS-style)
    edges = strong;
    [wy, wx] = find(weak);
    changed = true;
    while changed
        changed = false;
        for k = 1:length(wy)
            y = wy(k); x = wx(k);
            if ~edges(y, x)
                % Check 8-connected neighborhood for a strong edge
                neighbors = edges(max(1,y-1):min(H,y+1), max(1,x-1):min(W,x+1));
                if any(neighbors(:))
                    edges(y, x) = true;
                    changed = true;
                end
            end
        end
    end
end


function face_roi = detect_and_crop_face(img, target_size)
% DETECT_AND_CROP_FACE  Detect primary face and return normalized ROI.
%
%   Uses MATLAB's built-in vision.CascadeObjectDetector (Viola-Jones)
%   for Haar Cascade face detection. Falls back to center-crop if
%   the Computer Vision Toolbox is unavailable.
%
%   After detection, applies:
%       1. Grayscale conversion
%       2. Histogram equalization (CLAHE)
%       3. Gaussian blur (noise reduction)
%       4. Resize to target_size
%
%   Args:
%       img         : Input RGB or grayscale image (uint8, H×W or H×W×3)
%       target_size : [height, width] of output ROI (default [100,100])
%   Returns:
%       face_roi    : Normalized face ROI (uint8, target_size)
%                     or [] if no face detected

    if nargin < 2, target_size = [100, 100]; end

    gray = convert_to_grayscale(img);

    % Try Viola-Jones face detector (requires Computer Vision Toolbox)
    try
        detector = vision.CascadeObjectDetector();
        bbox = step(detector, img);
        if isempty(bbox)
            face_roi = [];
            fprintf('[PREPROCESS] No face detected.\n');
```

```
            return;
        end
        % Take the largest detected face by area
        [~, idx] = max(bbox(:,3) .* bbox(:,4));
        x = bbox(idx,1); y = bbox(idx,2);
        w = bbox(idx,3); h = bbox(idx,4);

        % Canonical inner face crop (focus on eyes, nose, mouth; ignore hair/collar variations)
        cy = max(1, floor(y + 0.10 * h));
        ch = min(size(gray, 1) - cy + 1, floor(0.82 * h));
        cx = max(1, floor(x + 0.08 * w));
        cw = min(size(gray, 2) - cx + 1, floor(0.84 * w));
        roi = gray(cy:cy+ch-1, cx:cx+cw-1);

    catch
        % Fallback: center crop (no Computer Vision Toolbox)
        fprintf('[PREPROCESS] CascadeObjectDetector unavailable.\n');
        fprintf('[PREPROCESS] Using center-crop as face ROI.\n');
        [H, W] = size(gray);
        margin_y = floor(H * 0.15);
        margin_x = floor(W * 0.15);
        roi = gray(margin_y+1:H-margin_y, margin_x+1:W-margin_x);
    end

    % Normalization pipeline
    face_roi = imresize(roi, target_size);      % Resize to fixed size
    face_roi = homomorphic_filter(face_roi);    % Frequency-domain illumination decoupling
    face_roi = histogram_equalization(face_roi);% Illumination normalization
    face_roi = gaussian_blur(face_roi, 0.8, 3); % Noise reduction
end


%% ========================================================================
%%  MODULE 2: FEATURE EXTRACTION & TEMPLATE GENERATION
%% ========================================================================

function hist_lbp = compute_lbp_histogram(gray, num_bins)
% COMPUTE_LBP_HISTOGRAM  Compute LBP texture descriptor.
%
%   Local Binary Pattern (LBP) encodes local texture by comparing each
%   pixel to its 8 neighbors. If neighbor ≥ center → bit=1, else bit=0.
%   The 8-bit result (0-255) is the LBP code for that pixel.
%
%   The normalized histogram of all LBP codes captures the distribution
%   of micro-textures across the face image.
%
%   Algorithm (8-neighbor, radius=1):
%       pattern = 0
%       For k = 0..7 (neighbors in clockwise order):
%           if I(neighbor_k) >= I(center): pattern |= (1 << k)
%       LBP[pixel] = pattern
%
%   Args:
%       gray     : Grayscale face ROI (uint8, H×W)
%       num_bins : Number of histogram bins (default 256)
%   Returns:
%       hist_lbp : Normalized LBP histogram (double, 1×num_bins)

    if nargin < 2, num_bins = 256; end

    g = double(gray);
    [H, W] = size(g);
    lbp = zeros(H, W, 'uint8');

    % 8-neighbor comparison (vectorized for speed)
    % Neighbor bit positions (clockwise from top-left):
    % bit0:top-left  bit1:top  bit2:top-right  bit3:right
    % bit4:bot-right bit5:bot  bit6:bot-left   bit7:left
    c = g(2:H-1, 2:W-1);  % Center pixels

    lbp(2:H-1,2:W-1) = ...
        uint8(g(1:H-2,1:W-2) >= c) * 1   + ...
        uint8(g(1:H-2,2:W-1) >= c) * 2   + ...
        uint8(g(1:H-2,3:W  ) >= c) * 4   + ...
        uint8(g(2:H-1,3:W  ) >= c) * 8   + ...
        uint8(g(3:H  ,3:W  ) >= c) * 16  + ...
        uint8(g(3:H  ,2:W-1) >= c) * 32  + ...
        uint8(g(3:H  ,1:W-2) >= c) * 64  + ...
        uint8(g(2:H-1,1:W-2) >= c) * 128;

    % Compute normalized histogram
    hist_lbp = histcounts(double(lbp(:)), 0:num_bins) / numel(lbp);
end


function hist_grad = compute_gradient_histogram(gray, num_bins)
% COMPUTE_GRADIENT_HISTOGRAM  Sobel gradient orientation histogram.
%
%   Computes the magnitude-weighted histogram of gradient orientations.
%   Captures the distribution of edge directions — encodes global facial
%   shape (contours of jaw, nose, eyebrows).
%
%   Algorithm:
%       1. Compute Sobel Gx, Gy
%       2. M = sqrt(Gx^2 + Gy^2)
%       3. θ = atan2d(Gy, Gx) + 360 (to map to [0°, 360°))
%       4. hist[bin] += M  (magnitude-weighted)
```

```
%
%   Args:
%       gray     : Grayscale face ROI (uint8, H×W)
%       num_bins : Orientation bins (default 36 → 10° per bin)
%   Returns:
%       hist_grad: Normalized gradient histogram (double, 1×num_bins)

    if nargin < 2, num_bins = 36; end

    [~, gx, gy] = sobel_edges(gray);
    magnitude   = sqrt(gx.^2 + gy.^2);
    orientation = mod(atan2d(gy, gx) + 360, 360);  % [0°, 360°)

    % Magnitude-weighted histogram
    edges = linspace(0, 360, num_bins + 1);
    hist_grad = zeros(1, num_bins);
    for b = 1:num_bins
        mask = orientation >= edges(b) & orientation < edges(b+1);
        hist_grad(b) = sum(magnitude(mask));
    end
    total = sum(hist_grad) + 1e-10;
    hist_grad = hist_grad / total;
end


function hog_feat = compute_hog(gray, cell_size, num_bins)
% COMPUTE_HOG  Histogram of Oriented Gradients (HOG) descriptor.
%
%   Divides the image into cell_size × cell_size pixel cells.
%   For each cell, computes an unsigned gradient orientation histogram
%   (0°–180°, num_bins bins). Normalizes overlapping 2×2 cell blocks
%   using L2 normalization for illumination robustness.
%
%   The concatenated block histograms form the HOG descriptor — a compact
%   and discriminative representation of local facial structure.
%
%   Args:
%       gray      : Grayscale face ROI (uint8, H×W)
%       cell_size : Pixel dimension of each cell (default 8)
%       num_bins  : Orientation bins per cell (default 9)
%   Returns:
%       hog_feat  : HOG feature vector (double, 1×N)

    if nargin < 2, cell_size = 8; end
    if nargin < 3, num_bins  = 9; end

    [H, W] = size(gray);
    ncx = floor(W / cell_size);
    ncy = floor(H / cell_size);

    % Compute gradients
    [~, gx, gy] = sobel_edges(gray);
    magnitude   = sqrt(gx.^2 + gy.^2);
    % Unsigned orientation [0°, 180°)
    orientation = mod(atan2d(abs(gy), abs(gx) + 1e-10), 180);

    % Per-cell gradient histogram
    cell_hists = zeros(ncy, ncx, num_bins);
    bin_width  = 180 / num_bins;

    for cy = 1:ncy
        for cx = 1:ncx
            y0 = (cy-1)*cell_size + 1;
            y1 = cy * cell_size;
            x0 = (cx-1)*cell_size + 1;
            x1 = cx * cell_size;

            cell_mag = magnitude(y0:y1, x0:x1);
            cell_ori = orientation(y0:y1, x0:x1);

            for b = 1:num_bins
                lo = (b-1)*bin_width;
                hi = b * bin_width;
                mask = cell_ori >= lo & cell_ori < hi;
                cell_hists(cy, cx, b) = sum(cell_mag(mask));
            end
        end
    end

    % Block normalization (2×2 cells per block, stride=1)
    block_size = 2;
    hog_feat = [];
    for by = 1:ncy-block_size+1
        for bx = 1:ncx-block_size+1
            block = cell_hists(by:by+block_size-1, bx:bx+block_size-1, :);
            block_vec = block(:)';
            norm = sqrt(sum(block_vec.^2)) + 1e-10;

% ... [Database file I/O omitted for brevity; core distance metrics shown below] ...

    end
    fprintf('\n%s\n', repmat('=', 1, 60));
    fprintf('%-6s  %-20s  %-22s  %s\n', 'ID', 'Name', 'Enrolled At', 'Tmpl Len');
    fprintf('%s\n', repmat('=', 1, 60));
    for i = 1:length(db.users)
        u = db.users(i);
        fprintf('%-6d  %-20s  %-22s  %d\n', ...
```

```
                u.id, u.name, u.enrolled_at, length(u.template));
    end
    fprintf('%s\n', repmat('=', 1, 60));
    fprintf('Total: %d user(s)\n\n', length(db.users));
end


function db = DB_delete_user(db, user_id)
% DB_DELETE_USER  Remove a user from the database by ID.
%
%   Args:
%       db      : Database struct
%       user_id : Integer ID of the user to delete
%   Returns:
%       db      : Updated database struct

    idx = find([db.users.id] == user_id);
    if isempty(idx)
        fprintf('[DB] User ID=%d not found.\n', user_id);
    else
        db.users(idx) = [];
        db = DB_save(db);
        fprintf('[DB] Deleted user ID=%d.\n', user_id);
    end
end


%% ========================================================================
%%  MODULE 4: MATCHING & IDENTIFICATION
%% ========================================================================

function dist = euclidean_distance(vec_a, vec_b)
% EUCLIDEAN_DISTANCE  Compute L2 (Euclidean) distance between two vectors.
%
%   Definition: d(a,b) = sqrt(sum((a_i - b_i)^2))
%
%   This is the primary metric for template matching. Smaller values
%   indicate more similar faces.
%
%   Args:
%       vec_a : Query feature vector (double, 1×N)
%       vec_b : Stored feature vector (double, 1×N)
%   Returns:
%       dist  : Scalar Euclidean distance

    dist = sqrt(sum((double(vec_a) - double(vec_b)).^2));
end


function sim = cosine_similarity(vec_a, vec_b)
% COSINE_SIMILARITY  Compute cosine similarity between two vectors.
%
%   Definition: sim(a,b) = (a · b) / (||a|| × ||b||)
%
%   Range: [0,1] for non-negative features. Higher = more similar.
%   Invariant to vector magnitude (illumination scale changes).
%
%   Args:
%       vec_a : Query feature vector (double, 1×N)
%       vec_b : Stored feature vector (double, 1×N)
%   Returns:
%       sim   : Cosine similarity score in [0,1]
```